

KOSTANAY, Kazakhstan

WMO: 289520

Lat: 53.2369N

Lon: 63.7408E

Elev: 558

StdP: 14.40

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-20.0	-32.0	0.9	-25.5	-25.7	1.3	-19.5	27.9	20.3	25.5	18.5	5.0	20	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	91.3	65.3	87.9	64.9	84.4	64.2	69.4	83.1	67.9	81.7	66.4	79.8	9.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.7	74.1	63.0	88.0	72.7	61.2	82.5	71.7	33.8	83.1	32.5	81.8	31.3	79.6	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	22.4	19.9	-31.7	96.3	6.4	3.4	-36.3	98.8	-40.0	100.7	-43.6	102.6	-48.3	105.1
			-32.0	73.2	6.3	2.7	-36.5	75.1	-40.1	76.7	-43.6	78.2	-48.2	80.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	3.0	5.6	20.7	42.5	57.1	66.3	68.6	66.1	54.7	40.9	23.2	8.9
	DBStd	25.95	12.65	12.60	10.45	10.60	8.90	7.77	6.68	7.81	8.70	8.80	12.44	12.91
	HDD50	6346	1458	1244	907	268	35	2	0	1	48	305	805	1273
	HDD65	10173	1923	1664	1372	674	274	76	39	84	326	748	1255	1738
	CDD50	2076	0	0	0	44	253	492	577	500	188	22	0	0
	CDD65	429	0	0	0	0	28	116	152	117	16	0	0	0
	CDH74	5181	0	0	0	25	468	1437	1661	1329	257	4	0	0
	CDH80	1912	0	0	0	3	125	553	620	523	88	0	0	0
Wind	WSAvg	8.9	8.1	8.8	10.1	10.2	9.9	9.0	8.3	8.1	8.3	8.7	9.0	8.7
Precipitation	PrecAvg	13.1	0.7	0.6	0.6	0.9	1.2	1.5	2.2	1.4	1.0	1.1	0.9	0.9
	PrecMax	18.7	1.6	1.4	1.5	2.6	3.6	4.1	5.9	4.0	3.0	4.3	2.0	2.3
	PrecMin	8.7	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.1	0.0	0.0	0.0
	PrecStd	2.7	0.4	0.4	0.4	0.6	0.8	1.1	1.5	0.9	0.8	0.8	0.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	35.1	51.6	77.2	87.9	93.6	96.5	95.1	89.7	71.8	50.3	35.9
		MCWB	32.8	32.9	44.0	55.3	62.8	65.3	66.0	66.2	61.5	53.4	45.5	33.2
	2%	DB	29.6	32.4	40.6	71.4	83.8	89.9	90.2	90.0	80.6	65.0	44.6	33.2
		MCWB	28.7	31.1	36.7	53.7	61.2	64.8	66.0	66.1	58.1	51.4	41.3	31.8
	5%	DB	26.4	29.0	37.0	66.0	79.4	86.4	86.3	85.9	75.3	60.0	41.1	30.2
		MCWB	25.3	27.9	34.5	51.4	59.1	64.3	65.7	65.6	57.3	48.5	38.3	28.9
	10%	DB	22.7	24.7	34.7	60.5	75.0	82.5	82.7	81.7	70.1	55.0	37.8	26.8
		MCWB	21.7	23.4	32.8	48.9	57.2	62.9	65.0	63.9	55.4	46.6	35.4	25.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	33.3	44.0	58.2	66.5	69.9	72.0	70.9	64.8	55.5	46.0	33.9
		MCDB	32.9	34.2	51.2	72.4	82.2	84.0	83.9	85.7	81.5	67.6	49.0	35.1
	2%	WB	28.5	31.0	37.0	54.9	62.9	68.0	69.8	68.9	60.9	52.4	41.5	31.5
		MCDB	29.3	32.2	39.7	68.1	79.3	84.1	81.9	84.5	75.5	62.9	43.9	32.7
	5%	WB	25.4	27.9	34.7	52.4	60.6	66.3	68.2	67.1	58.6	49.6	38.5	28.8
		MCDB	26.3	29.0	36.5	63.8	75.6	81.3	80.1	81.4	72.1	58.4	40.7	30.0
	10%	WB	21.7	23.5	32.9	49.5	58.3	64.5	66.7	65.3	56.4	47.0	35.6	25.8
		MCDB	22.7	24.7	34.4	59.7	72.0	78.2	79.0	77.9	68.1	54.0	37.5	26.9
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.9	17.0	20.7	23.2	22.7	20.9	22.2	22.1	18.0	13.7	15.5
		MCDBR	14.8	15.5	15.1	29.6	30.0	28.4	26.7	28.4	30.2	27.3	15.1	14.1
		MCWBR	14.3	14.6	11.8	16.0	14.1	11.3	10.7	11.9	13.8	15.9	11.9	13.3
	5% WB	MCDBR	14.5	15.3	14.4	26.9	27.0	24.7	22.5	24.9	26.1	25.1	14.4	14.5
		MCWBR	14.3	14.6	11.8	16.0	14.0	11.6	10.8	11.8	13.6	15.7	11.9	14.0
Clear-Sky Solar Irradiance	taub		0.289	0.321	0.372	0.409	0.408	0.404	0.420	0.433	0.380	0.353	0.320	0.289
	taud		2.080	2.026	1.978	2.134	2.226	2.313	2.265	2.202	2.302	2.283	2.194	2.067
	Ebn at Noon		207	237	249	259	266	269	261	249	249	228	195	179
	Edh at Noon		23	34	44	43	42	38	40	40	32	26	21	19
All-Sky Solar Radiation	RadAvg		364	724	1201	1497	1820	1974	1857	1534	1104	590	295	235
	RadStd		28	32	109	156	160	151	129	148	84	84	29	41

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page